IEEE ZAGAZIG Student Branch

IEEE COMPUTER SOCIETY ZAGAZIG STUDENT CHAPTER

@Backend:Phase1

Task 5

/DBMS-MySQL

This is the final task of MySQL, and next is the project of the 1st phase.

/Feedback

$ Task_4_Feedback

/Resources

$ Attend the online session(there is a warning if you don't attend it).

$ from Baraa Course, watch these videos:

> Window Functions Basics

Indexes

> SQL Indexes

> SQL Unique & Filtered Indexes

> How to choose the right index

Transactions

> Database Transactions

$ Other Useful resources along this phase:

> Baraa Course

> Eng-Ramy ITI Course

> DR-Amr Internal DBs Course

/TODO

$ Part A: LeetCode Problems

Create a file named Task_5_Solutions.sql and paste your SQL code for each problem there.

- > Duplicate Emails
- > Delete Duplicate Emails
- > Nth Highest Salary
- > Rank Scores
- > Department Highest Salary

$ Part B: Research Questions

Answer the following questions in the file named **research.md**.

Window Functions vs GROUP BY:

- Both GROUP BY and window functions perform aggregations. What is the fundamental difference regarding the "granularity" (level of detail) in the output?

Clustered vs Non-Clustered Indexes:

- Based on the "Clustered vs Nonclustered" video, explain the difference between the **Leaf Nodes** of a Clustered Index's B-Tree versus a Non-Clustered Index.

- *Why are you only allowed to have exactly **one clustered index** per table?*

Filtered & Unique Indexes:

- *What is a "Filtered Index" and why is it useful? (Mention the impact on storage and query performance).*
- *If you put a "Unique Index" on an Email column, how does it physically slow down INSERT statements while speeding up SELECT statements?*

Choosing the Right Index:

- *You have a temporary "Staging Table" where you insert millions of rows very quickly, read them once, and then delete them. According to the "How to Choose" video, should you use a Clustered Index, a Non-Clustered Index, or a Heap Structure? Why?*

Database Transactions (ACID):

- *Explain the "All or Nothing" concept (Atomicity). What disastrous scenario happens if a partial failure occurs without using a Transaction?*

/Submission Guidelines

> Inside your IEEE-ZSB-Backend-26 repository, inside the MySQL folder, create a new folder named Task_5 and put your files there:

```
MySQL\
    Task_5\
        Task_5_Solutions.sql
```

research.md

HTML\

. . .

/How-To-Submit

$ After you finish all of that, push your files to the repo as explained.

$ Submit the GitHub repository link on the task chat on **Discord.**

/Deadline

$ Saturday, February 28 at **11:59 PM**

Wish you the best.❤️